

Individual Final Report

1. Introduction

This report summarizes my work on stance detection, the models I developed, and my final analysis. Throughout my project, I focused on classifying tweets as "favor", "against", or "neither" about specific targets using the SemEval-2016 Task 6: Detecting Stance in Tweets dataset. While conducting this work, I based my work on the article "Detecting Stance in Tweets Using Shallow Features and GloVe Vectors for Word Representation" by Henrik Böhler, Petter Fagerlund Asla, and others. In the article, high performance was achieved by combining GloVe vectors and n-gram based features. My work aimed to improve the performance by replacing this basic approach with modern methods (such as BERT).

2. Project Scope and Approaches

During my project, I developed three different models and a baseline model. My models aimed to extend the system in the article by offering different approaches to the stance detection problem.

3. Models Used

- Model 1: Direct Naive Bayes + BERT

Approach: Stance detection was performed with the Naive Bayes model by combining all target categories. Tweets were converted to BERT-based deep contextual vectors, then prediction was made with the Multinomial Naive Bayes classifier.

Result: This model was especially successful in the AGAINST class, but showed low performance in the FAVOR and NEITHER classes. It provided limited success because it ignored the category-specific context used in the article.

F1-Score (Stance): 0.4685 (Test Set).

- Model 2: Category-Specific Naive Bayes + BERT

Approach: Separate Naive Bayes models were created for each target category and category-specific contexts were better captured. This model used category-specific BERT vectors for stance detection.

Result: AGAINST class showed high performance in most categories. It gave better results compared to Model 1.

F1-Score (Stance): 0.5400 (Test Set).

- Model 3: Category + Stance (Logistic Regression + Naive Bayes)

Approach: A two-stage process was implemented. In the first stage, category prediction was made using TF-IDF based Logistic Regression. In the second stage, separate stance detection models were created for each category.

Result: This model provided high accuracy in category prediction. However, the low performance in FAVOR and NEITHER classes in stance detection limited the overall results.

F1-Score (Stance): 0.5400 (Test Set).

4. Performance Comparison

Model	Test Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-Score
Model 1	0.5632	0.5343	0.4660	0.4685
Model 2	0.57	0.55	0.55	0.54
Model 3	0.57	0.55	0.55	0.54
Article	0.6247	0.5750	0.5287	0.6247